

Packet Tracer - Examine the ARP Table

Addressing Table

Device	Interface	MAC Address	Switch Interface
Router0	Gg0/0	0001.6458.2501	G0/1
	S0/0/0	N/A	N/A
Router1	G0/0	00E0.F7B1.8901	G0/1
	S0/0/0	N/A	N/A
10.10.10.2	Wireless	0060.2F84.4AB6	F0/2
10.10.10.3	Wireless	0060.4706.572B	F0/2
172.16.31.2	F0	000C.85CC.1DA7	F0/1
172.16.31.3	F0	0060.7036.2849	F0/2
172.16.31.4	G0	0002.1640.8D75	F0/3

Objectives

Part 1: Examine an ARP Request

Part 2: Examine a Switch MAC Address Table

Part 3: Examine the ARP Process in Remote Communications

Background

This activity is optimized for viewing PDUs. The devices are already configured. You will gather PDU information in simulation mode and answer a series of questions about the data you collect.

Instructions

Part 1: Examine an ARP Request

Step 1: Generate ARP requests by pinging 172.16.31.3 from 172.16.31.2.

- Click **172.16.31.2** and open the **Command Prompt**.
- Enter the **arp -d** command to clear the ARP table.
- Enter **Simulation** mode and enter the command **ping 172.16.31.3**. Two PDUs will be generated. The **ping** command cannot complete the ICMP packet without knowing the MAC address of the destination. So the computer sends an ARP broadcast frame to find the MAC address of the destination.
- Click **Capture/Forward** once. The ARP PDU moves **Switch1** while the ICMP PDU disappears, waiting for the ARP reply. Open the PDU and record the destination MAC address.

Is this address listed in the table above?

FFFF.FFFF.FFFF, and no, it is not listed in the table above.

- e. Click **Capture/Forward** to move the PDU to the next device.

How many copies of the PDU did **Switch1** make?

3 copies

What is the IP address of the device that accepted the PDU?

172.16.31.3

- f. Open the PDU and examine Layer 2.

What happened to the source and destination MAC addresses?

The destination MAC address is now the original device we pinged from (172.16.31.2), and the src MAC address is now the device we wanted to ping (172.16.31.3)

- g. Click **Capture/Forward** until the PDU returns to **172.16.31.2**.

How many copies of the PDU did the switch make during the ARP reply?

Only one, because the reply is a unicast back to the original device with the new MAC information.

Step 2: Examine the ARP table.

- a. Note that the ICMP packet reappears. Open the PDU and examine the MAC addresses.

Do the MAC addresses of the source and destination align with their IP addresses?

Yes, it now contains the MAC address of the device with IP 172.16.31.3, and the src MAC address is the device at IP 172.16.31.2.

- b. Switch back to **Realtime** and the ping completes.

- c. Click **172.16.31.2** and enter the **arp -a** command.

To what IP address does the MAC address entry correspond?

172.16.31.3, with the corresponding MAC address 0060.7036.2849.

In general, when does an end device issue an ARP request?

When the end device only has the IP address of the device they are trying to connect with. It needs the MAC address in order to send the packets to said device.

Part 2: Examine a Switch MAC Address Table

Step 1: Generate additional traffic to populate the switch MAC address table.

- From **172.16.31.2**, enter the ping **172.16.31.4** command.
- Click **10.10.10.2** and open the **Command Prompt**.
- Enter the **ping 10.10.10.3** command.

How many replies were sent and received?

4 were sent and received

Step 2: Examine the MAC address table on the switches.

- a. Click **Switch1** and then the **CLI** tab. Enter the **show mac-address-table** command.

Do the entries correspond to those in the table above?

Yes, it contains the MAC addresses of the router, devices 172.16.31.4, and 172.16.31.2.

- b. Click **Switch0**, then the **CLI** tab. Enter the **show mac-address-table** command.

Do the entries correspond to those in the table above?

Yes, it contains the MAC addresses of the router, devices 10.10.10.2, and 10.10.10.3.

Why are two MAC addresses associated with one port?

Because they are both connected to the access point, which only connects to one port on the switch.

Part 3: Examine the ARP Process in Remote Communications

Step 1: Generate traffic to produce ARP traffic.

- Click **172.16.31.2** and open the **Command Prompt**.
- Enter the **ping 10.10.10.1** command.
- Type **arp -a**.

What is the IP address of the new ARP table entry?

It is listed as **172.16.31.1**.

- Enter **arp -d** to clear the ARP table and switch to **Simulation** mode.
- Repeat the ping to 10.10.10.1.

How many PDUs appear?

Two, since we cleared the ARP table.

- Click **Capture/Forward**. Click the PDU that is now at **Switch1**.

What is the target destination IP destination address of the ARP request?

172.16.31.1

- The destination IP address is not 10.10.10.1.

Why?

Since we can't find the address 10.10.10.1 in our ARP table, it might be outside the network, in which we would need to send the ARP request to the default gateway, which this may be the address for. The packet gets passed along to the router, which is the default gateway, before making it's way back to 172.16.31.2.

Step 2: Examine the ARP table on Router1.

- Switch to **Realtime** mode. Click **Router1** and then the **CLI** tab.
- Enter privileged EXEC mode and then the **show mac-address-table** command.

How many MAC addresses are in the table? Why?

The command is not the same as looking at the ARP table as we do with a switch
(<https://community.cisco.com/t5/routing-and-sd-wan/mac-address-table-in-ios-routers/td-p/2541350>)

- Enter the **show arp** command.

Is there an entry for **172.16.31.2**?

Yes

What happens to the first ping in a situation where the router responds to the ARP request?

It may register as a timed out request.